

Full Prophage Homology Survey

MASH-based Analysis of 132,393 Prophage Sequences from 26,000 E. coli Genomes

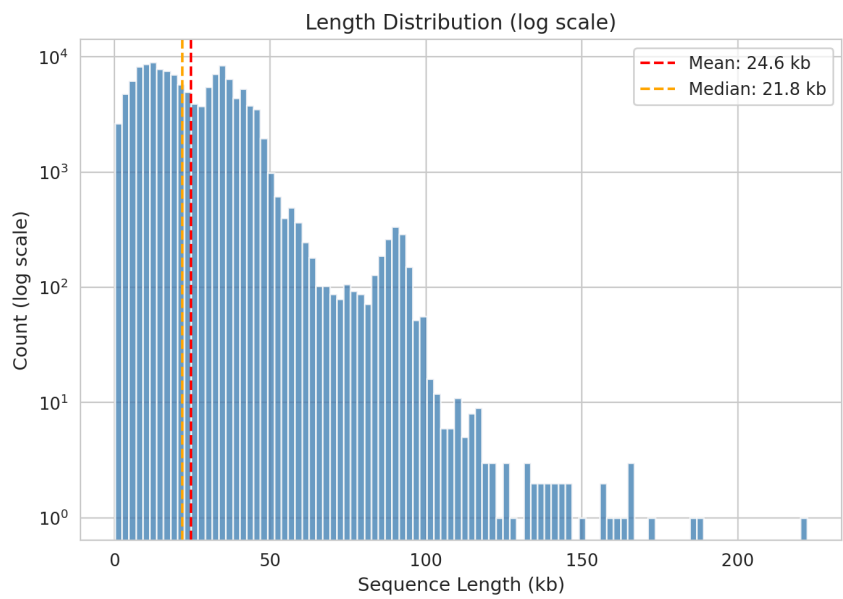
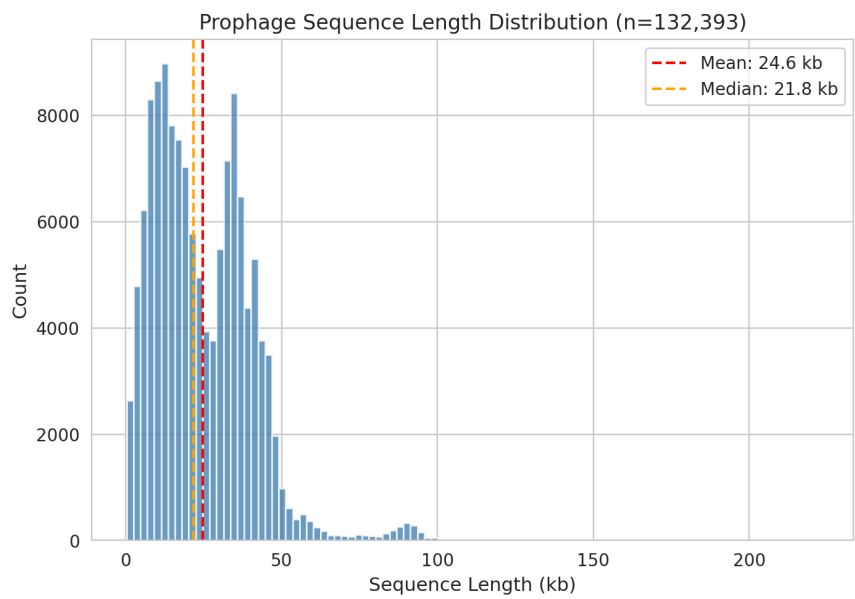
132,393
Total Sequences

8,187
Genomes

24.6 kb
Mean Length

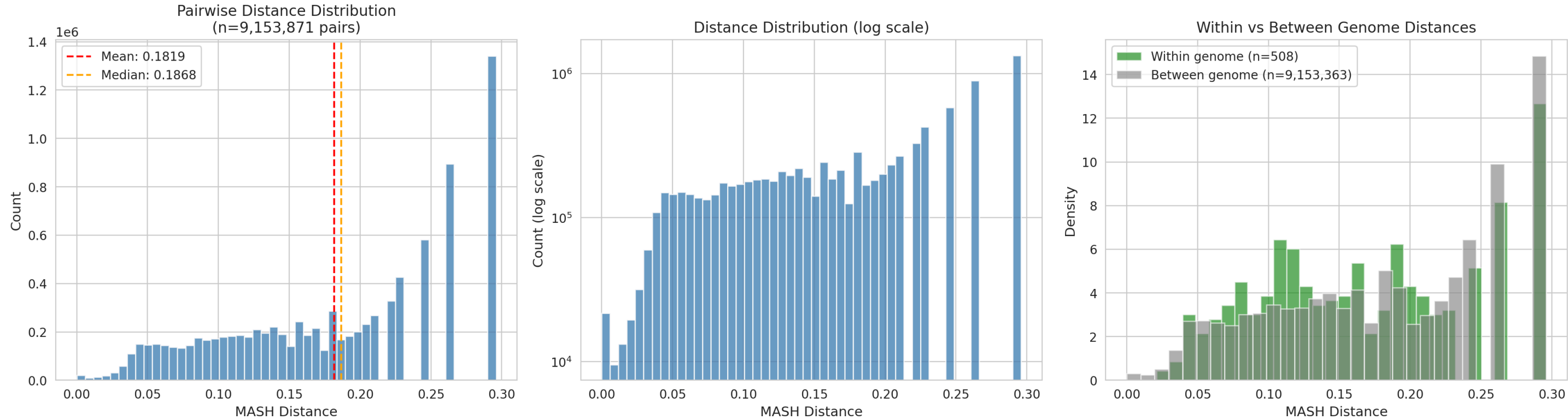
9,153,871
Similar Pairs (d<0.5)

1. Sequence Length Distribution



Metric	Value
Count	132,393
Mean	24.6 kb
Median	21.8 kb
Std Dev	15.9 kb
Min	0.3 kb
Max	222.3 kb
Q25	11.9 kb
Q75	35.2 kb

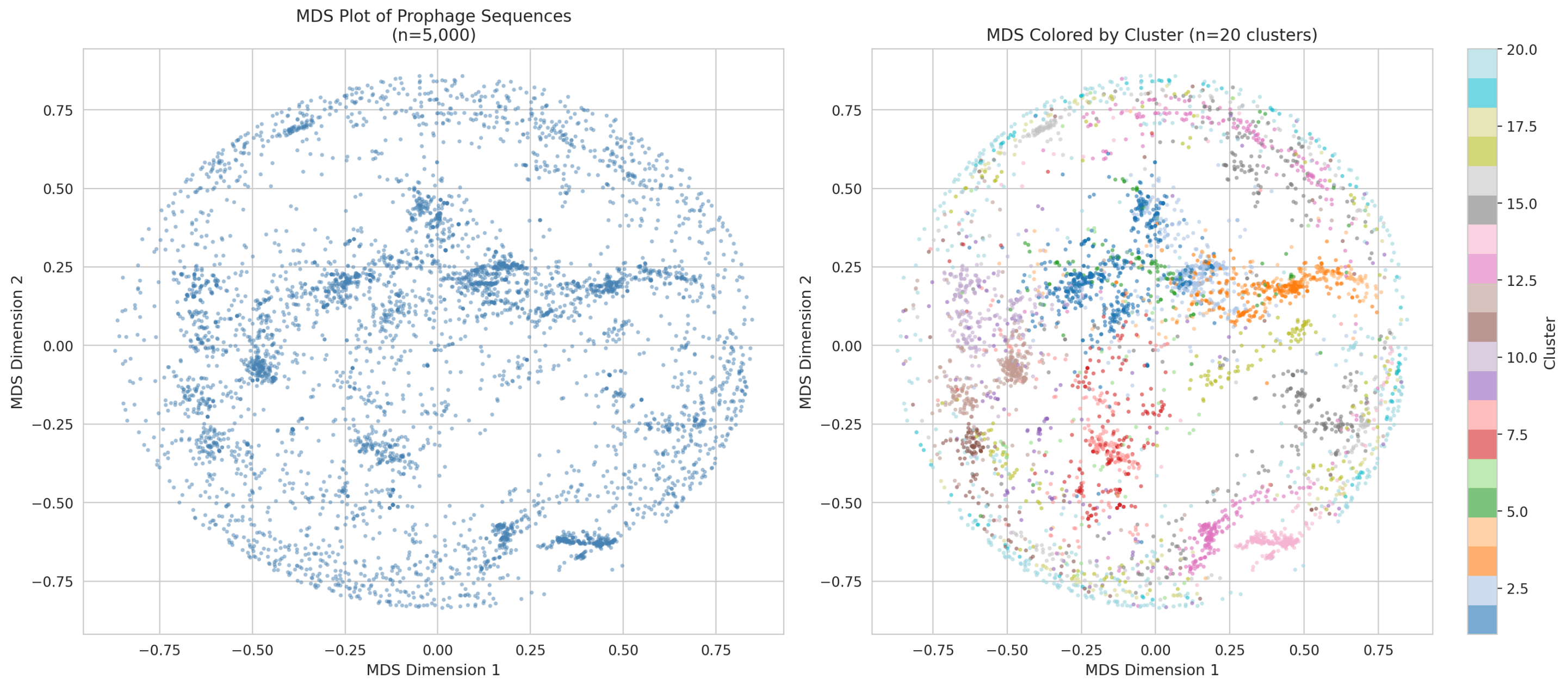
2. MASH Distance Distribution



Metric	Overall	Metric	Within Genome	Metric	Between Genome
Pairs	9,153,871	Pairs	508	Pairs	9,153,363
Mean	0.1819	Mean	0.1695	Mean	0.1819
Median	0.1868	Median	0.1647	Median	0.1868
Std	0.0797	Std	0.0761	Std	0.0797

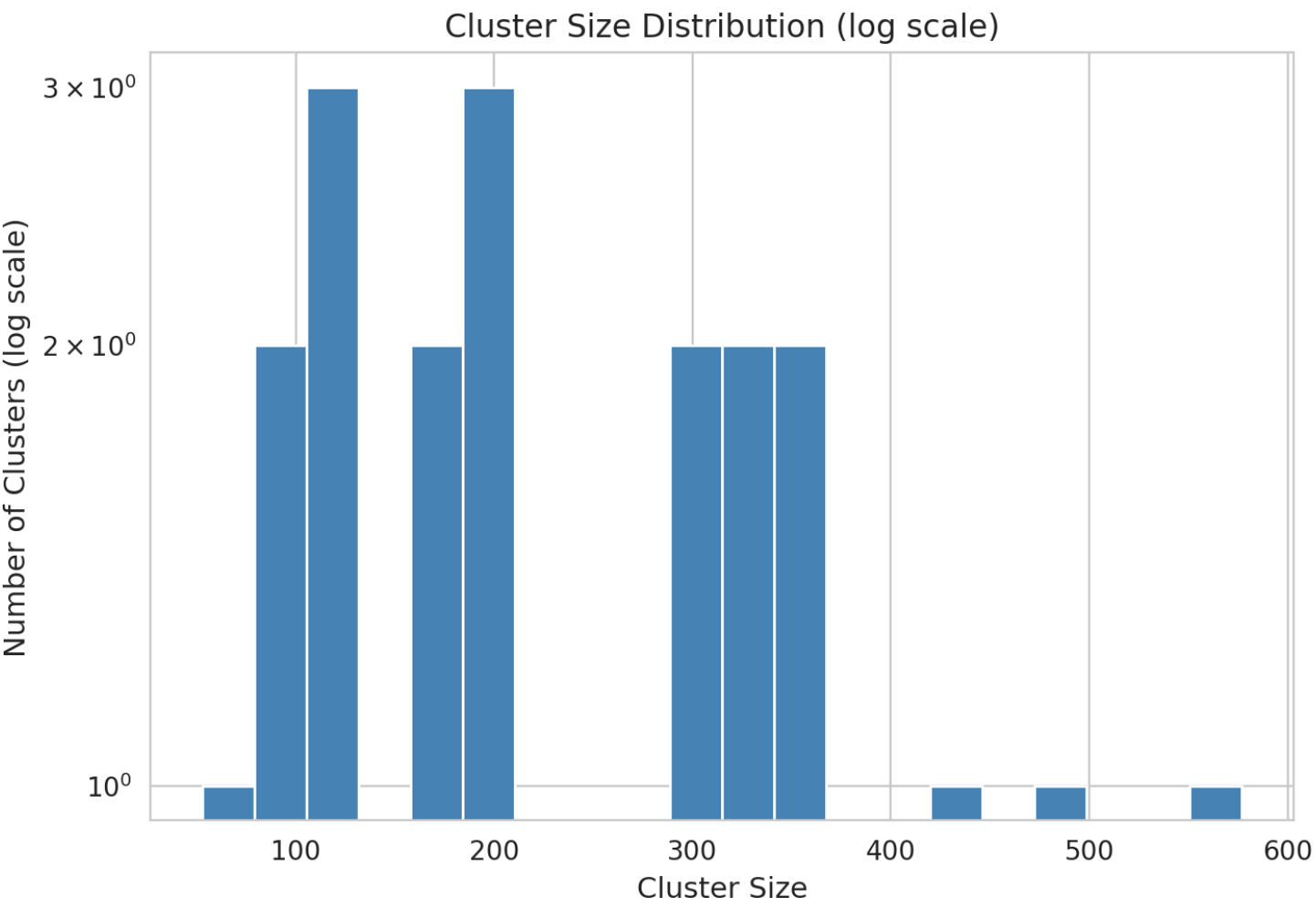
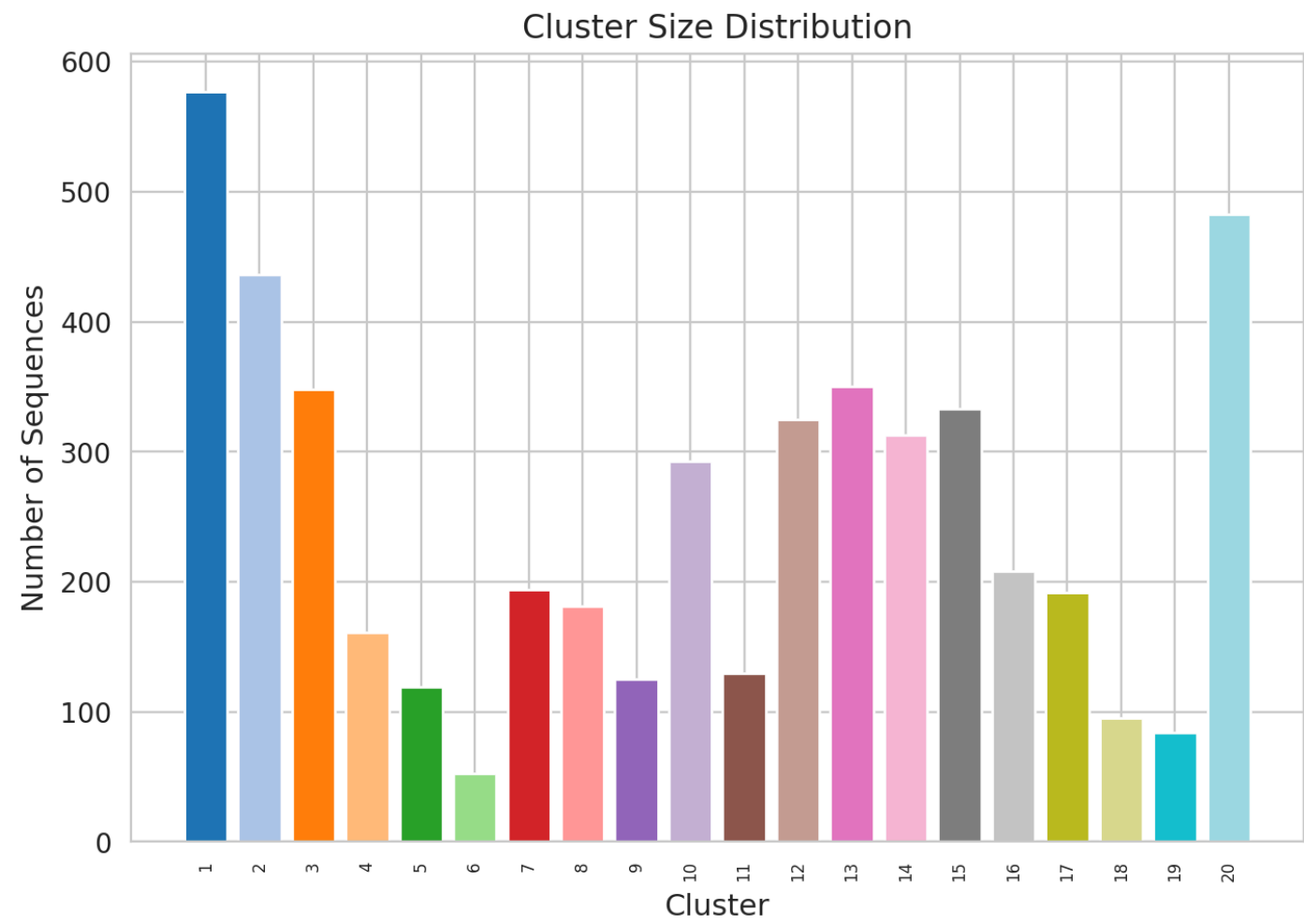
3. Multidimensional Scaling (MDS)

MDS stress: 1626006.7133

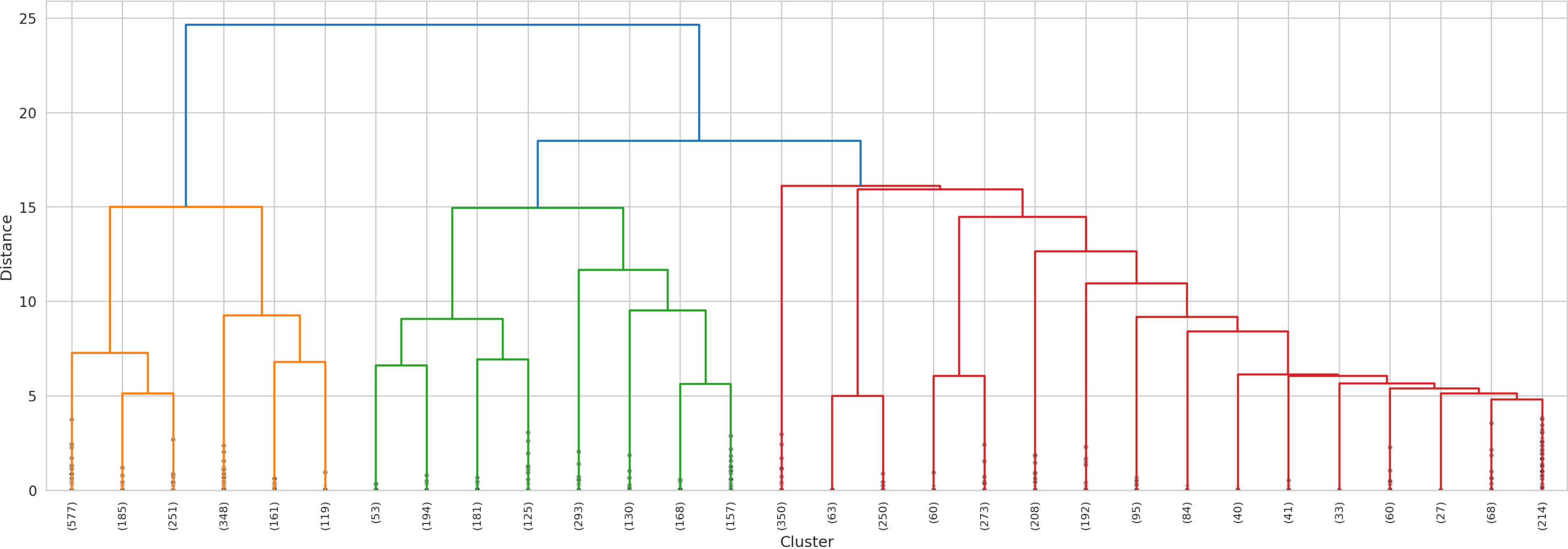


4. Hierarchical Clustering

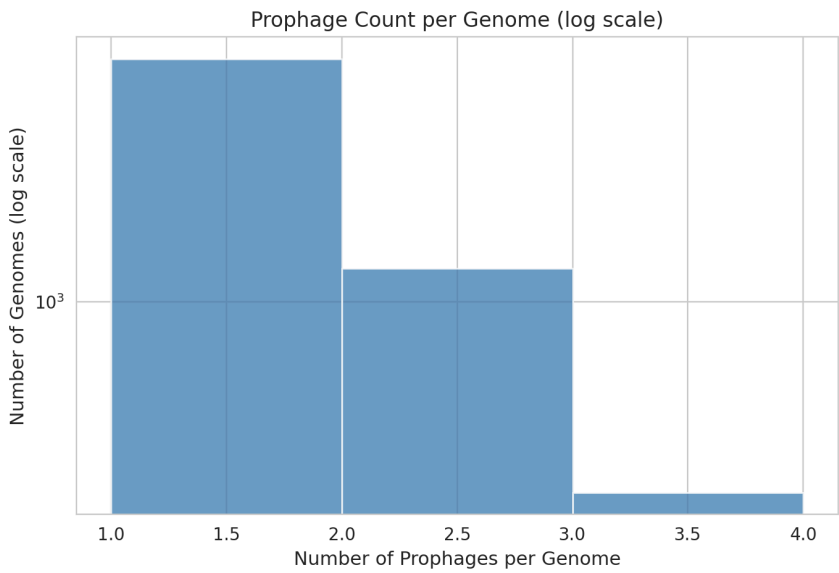
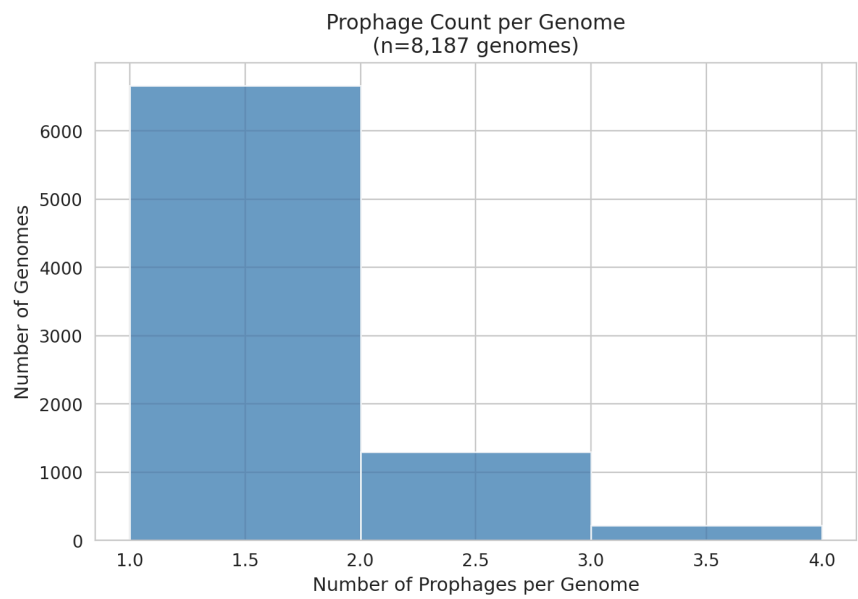
Optimal number of clusters: 20



Hierarchical Clustering Dendrogram (truncated)

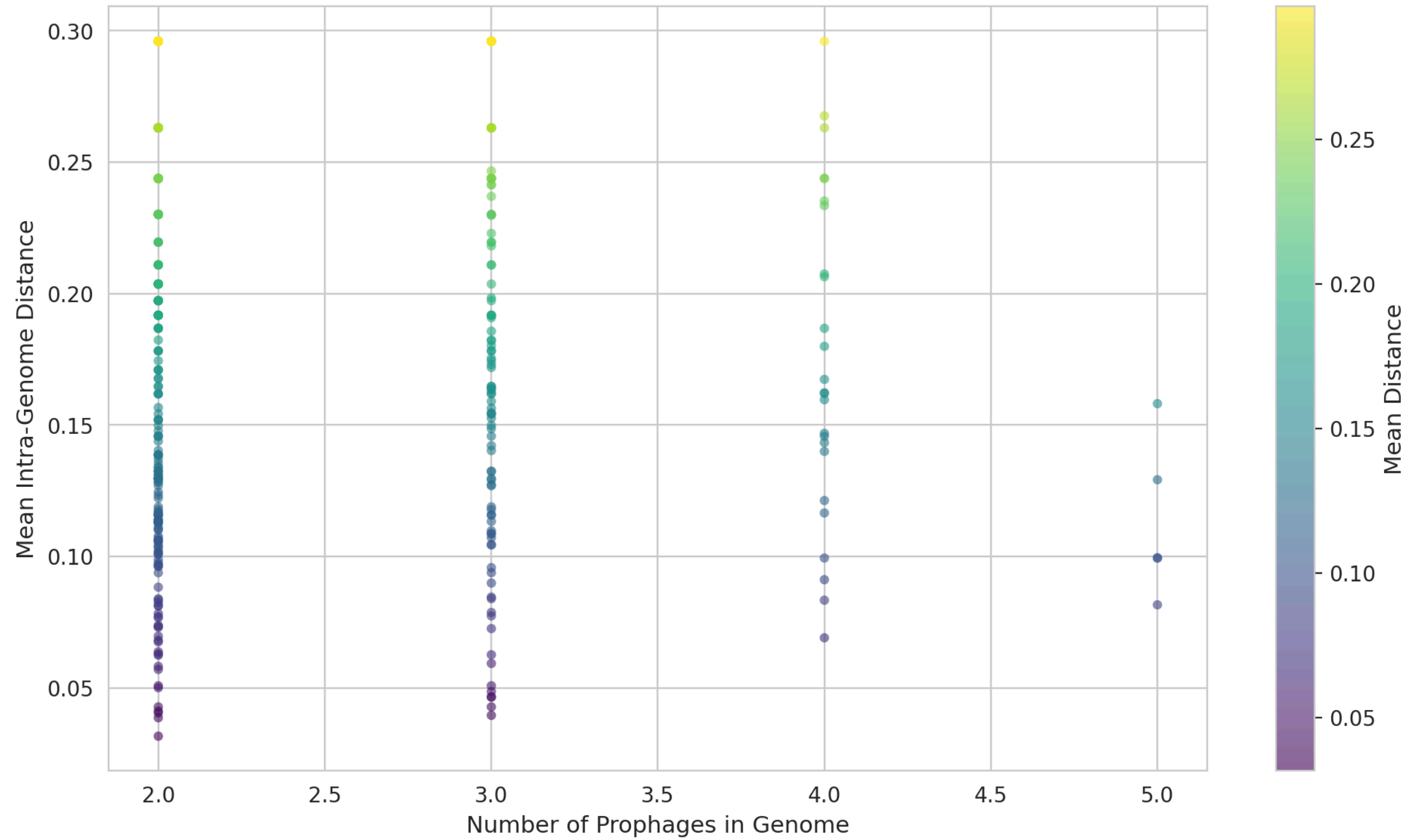


5. Genome-Level Prophage Diversity



Metric	Value
Genomes with prophages	8,187
Genomes with 2+ prophages	1,527
Max prophages/genome	5
Mean prophages/genome	1.2

Intra-Genome Prophage Diversity



6. Summary Statistics

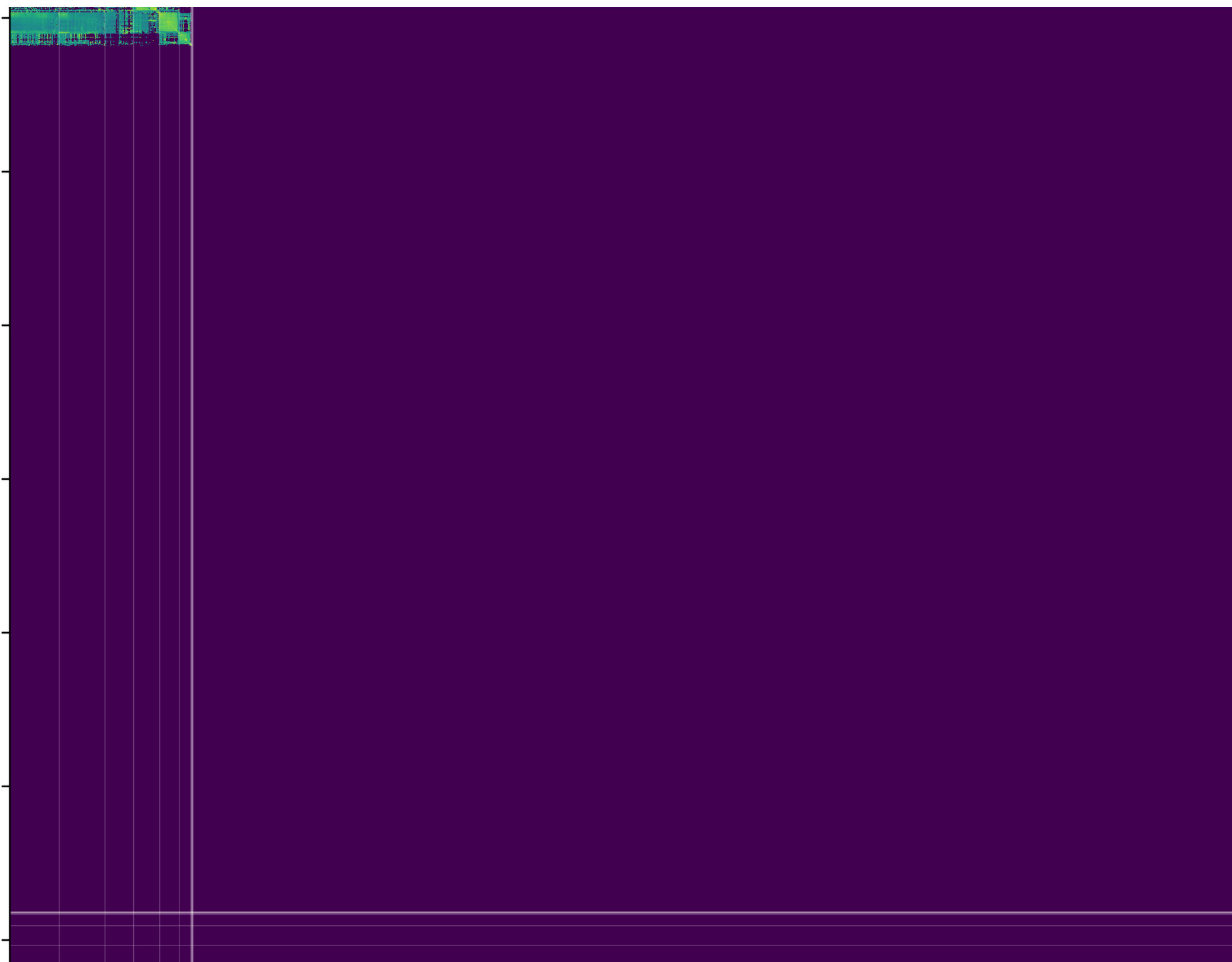
Metric	Value	Notes
Total sequences	132,393	Full prophage set
Total genomes	8,187	From 26k E. coli genomes
Mean sequence length	24.6 kb	Range: 0.3-222.3 kb
Similar pairs (d<0.5)	9,153,871	Out of 9e+09 total pairs
Mean distance (similar pairs)	0.1819	Only pairs with d<0.5
MDS stress	1626006.7133	Lower is better
Number of clusters	20	Hierarchical clustering
Genomes with 2+ prophages	1,527	Intra-genome diversity

7. Full-Resolution Heatmap (Community Detection)

Full 132k×132k clustered heatmap using Louvain community detection on all 36M similar pairs. The matrix is binned to 2000×2000 resolution. White lines show community boundaries. Dark regions indicate closely related prophage clusters. Of 132,393 sequences, 19,638 have at least one similar pair ($d < 0.5$) and form 12 communities; 112,755 are singletons with no similar pairs.

Sequence (ordered by community)

1750
1500
1250
1000
750
500
250



MASH Distance

0.4

0.3

0.2

0.1

8. Cross-Community Homology Matrix

Mean pairwise MASH distance between each pair of communities.
Values near 0 indicate closely related communities; values near 0.5 indicate distant relationships.
Diagonal is zero (within-community). Showing 12 connected communities;
112,755 singleton communities (no similar pairs) are omitted.

